DATA ITEM DESCRIPTION

Title: NUCLEAR SURVIVABILITY TEST REPORT

Number: DI-NUOR-80929B

Approval Date: 20130201

AMSC Number: F9320

Limitation:

DTIC Applicable: N/A

GIDEP Applicable:

Preparing Activity: 27 (AFNWC)

Applicable Forms: N/A

Use/Relationship: The Nuclear Survivability Test Report shall document the results of nuclear survivability test performed on systems, subsystems, structures, circuits, components, materials, and piece parts and provide the engineering evaluation of the system's nuclear survivability.

a. When a solicitation or contract contains a requirement to demonstrate compliance with nuclear survivability requirements, this Data Item Description (DID) shall be listed on the Contract Data Requirements List (DD Form 1423). The documents cited in this DID are to be used for guidance and are not mandatory for the CDRLs.

b. This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirements as delineated in the contract.

c. When this DID is imposed in a solicitation or contract, DI-NUOR-80928, *Nuclear Survivability Test Plan*, shall also be listed on the DD Form 1423.

(Copies of the DID are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Ave., Bldg 4D, Philadelphia PA 19111-5094.)

d. Air Force Instruction (AFI) 10-2607, *Air Force Chemical, Biological, Radiological, and Nuclear (CBRN) Survivability*, contains the AF policy applicable to nuclear survivability.

(Copies of the AF Instruction are available online at http://www.e-publishing.af.mil/.)

e. The DD Form 1423 shall also list the Defense Threat Reduction Information Analysis Center (DTRIAC), 1680 Texas St., SE, Kirtland AFB NM 87117-5669 or e-mail dtriac@dtra.mil, as a recipient for the report. DTRIAC serves as a DoD wide archive for nuclear survivability purposes.

f. A separate copy of the Special Data (see 3.4) shall be provided to DTRIAC (see e. above).

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions, shall be as specified in the solicitation or contract.

2. Format. The Nuclear Survivability Test Report shall be in contractor's format.

3. General: The Nuclear Survivability Test Report shall contain all the data and conclusions resulting from the test(s) delineated in the current revision of DI-NUOR-80928 and shall include the following information:

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3.1 Specific data described in the report.

a. Documentation with raw data of the pre- and post-exposure measurements on the piece parts, materials, components, circuits, structures, subassemblies, assemblies, and systems tests. There shall be particular emphasis on identifying and reporting the test results for the hardness critical items (HCI) or harness critical process (HCP) and the mission essential equipments.

b. The report shall document the test conditions for these measurements, including the ambient conditions, the instrumentation for the measurements, the monitor points, the time relation of the measurements with respect to the test level, modes and conditions of the test sample function during the test level and during the measurements, physical orientations of the test samples to the test environment (for system level tests, these include the Government and contractor furnished equipment), time after exposure that the measurements are made and the measurements of the simulated environments.

c. The report shall provide a statement of how the simulated environments were measured; where, with respect to the sample, the measurements were made and what devices or instruments were used.

d. A detailed description of the instrumentation calibration procedures and dosimetry shall be provided.

e. Detailed test data sheets for each test level with the concurrence signatures of the Government observers and the contractors shall be provided by the contractor in this report.

f. The report shall list the applicable methods of MIL-STD-750 and MIL-STD-883 methods used to design the tests on semiconductor piece parts.

(Copies of MIL-STDs-750 and -883 are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Oder Desk, 700 Robbins Ave., Bldg 4D, Philadelphia PA 19111-5094.)

3.2 System test operational data described in the report:

a. For system level tests, the report shall specify the system operational steps used to set up the system for tests.

b. The report shall detail the equipment post-exposure status and the procedures used to restore the system to full function after the exposure.

3.3 Data analysis described in the report:

a. The report shall provide an analysis for the acceptability of the test data.

b. As appropriate, a detailed description shall be provided of how the response information on piece parts, materials, components, circuits, structures, subassemblies and assemblies is analyzed to arrive at an evaluation of the system survivability, to include the operational impact of the system's nuclear response, error budgets and the specification for the allowable downtime.

c. Details shall be provided for the calculations and analysis for the interpolation or extrapolation of the test data to determine how the system responds to the specified criteria. Computer analysis codes used in the analysis shall be named and referenced. All configurations, vehicles, and enclosures for these equipments shall be included in the nuclear survivability evaluations.

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d. In the data analysis discussions, the assumptions, opinions and judgments made to support the conclusions shall be clearly identified.

3.4 Special data for electronic parts. The test report(s) shall provide the data listed below for the neutron, ionizing radiation dose and ionizing dose-rate responses of electronic piece parts. A copy of this data is to be provided in digital form to DTRIAC as specified (see e and f. above).

a. Description of device tested.

(1) Type number to include complete MIL-STD part designations, as appropriate (e.g., 2N2930 JTX, M38510/00101BAC), and generic or Joint Electron Device Engineering Council (JEDEC) equivalent number.

(2) For non-standard devices, the closest generic or JEDEC part number.

(3) Device class (for example, zener diode, low-power transistor, or linear or digital integrated circuit).

(4) Manufacturer.

(5) Device test sequence identification number.

(6) Manufacturer's lot number, if available, and lot type, i.e., wafer, diffusion or inspection.

(7) Date code or purchase date.

(8) Number of samples tested.

(9) Governing electrical specifications.

(10) Package type.

b. Neutron and ionizing radiation dose effects data.

(1) Pre-and post irradiation parameters measured, operating points (for example, h_{FE} with I_C and V_{CE} operating points), time between irradiation and measurement, condition of devices between irradiation and measurement (i.e., temperature, mobile bias fixture, time unbiased and applied voltages) and time between consecutive measurements.

(2) Test facility (e.g., reactor, White Sands Missile Range, Survivability and Vulnerability Directorate or Febatron 706.).

(3) Test date.

(4) Bias conditions applied during irradiation.

(5) Irradiation temperature and test temperature.

(6) Test data for each piece part at each radiation level.

(7) Ionizing radiation dose (rad(si)) or neutron fluence (n/cm^2 (1 MeV equivalent damage in Si)). For ionizing radiation dose measurements, the radiation type, dose rate and energy and the duration of the irradiation(s) must be specified.

(8) Applicable MIL-STD methods.

(9) Exposure circuitry and configuration, giving details of materials, distances, designs and applied voltages.

(10) Parameter and failure measurement apparatus and circuitry.

(11) Dosimetry methods and applicable ASTM standards.

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(12) Remarks/comments – relevant information to explain or clarify test conditions or data.

(13) Anomalous incidents or data.

(14) Post-irradiation effects response data for the ionizing radiation dose.

c. Ionizing dose-rate effects data.

(1) Same information as 3.4b, excluding items 7 and 14.

(2) Pulse with full width half maximum and pulse shape.

(3) Photocharge in coulombs or peak primary photocurrent in milliamperes.

(4) Cumulative dose in rad(Si) and dose rate in rad(Si)/second.

(5) As applicable, upset or latchup threshold data

d. Data Summary

(1) Calculated means and standard deviation for each lot sample.

(2) The aggregate means and standard deviations for the combined lots.

(3) The 90 percent probability, 90 percent confidence limit of response data.

e. General Information

(1) Name and address of company taking data.

(2) Person(s) responsible for tests and data records

(3) Phone number for point of contact to include area code and extension.

(4) Any identifier(s) that will assist in locating data such as lab book numbers.

4. End of DI-NUOR-80929B.

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